

Argument Principle.

Let  $f$  be a non-constant function, analytic at  $z_0$  with  $f(z_0) = 0$ .

Then, for some  $r > 0$ ,  $f$  have only  $z_0$  as  $z_0$  in  $B_r(z_0)$ . Thus,

$$f(z) = (z - z_0)^k \cdot F(z) \text{ for some } k \in \mathbb{N},$$

and analytic function  $F$  at  $z_0$  with  $F(z_0) \neq 0$ . Now,

$$f'(z) = k(z - z_0)^{k-1} F(z) + (z - z_0)^k \cdot F'(z),$$

$$\text{this implies } \frac{f'(z)}{f(z)} = \frac{k}{z - z_0} + \frac{F'(z)}{F(z)}.$$

Since  $F(z_0) \neq 0$ ,  $f'/f$  has simple pole at  $z_0$ , with residue  $k$ . So,

$$\int_C \frac{f'(z)}{f(z)} dz = 2\pi i \cdot k \text{ where } C \text{ is simple closed curve around } z_0, \\ \text{is contained in } B_r(z_0).$$

Now, let  $f$  has a pole of order  $n$  at  $z_0$ , then

$$f(z) = \frac{F(z)}{(z - z_0)^n}$$

where  $F$  is analytic at  $z_0$  and  $F(z_0) \neq 0$ . Then,

$$f'(z) = -n \cdot \frac{F(z)}{(z - z_0)^{n+1}} + \frac{F'(z)}{(z - z_0)^n},$$

$$\text{so therefore } \frac{f'(z)}{f(z)} = \frac{(z - z_0)^n}{F(z)} \cdot \left( -n \cdot \frac{F(z)}{(z - z_0)^{n+1}} + \frac{F'(z)}{(z - z_0)^n} \right) = \frac{-n}{z - z_0} + \frac{F'(z)}{F(z)}$$

$$\text{so, } \int_C \frac{f'(z)}{f(z)} dz = 2\pi i \cdot (-n).$$

Obtain the two results, we obtain the following:

Thm. Let  $f$  be analytic inside and on a simple closed contour  $C$  except for finite poles in  $C$ , and  $f(z) \neq 0$  on  $C$ . Then, counted with multiplicity, numbers of zeros  $N_f$  and poles  $P_f$ ,

$$\frac{1}{2\pi i} \cdot \int_C \frac{f'(z)}{f(z)} dz = N_f - P_f.$$

[More observation.

Suppose that  $f(z)$  is analytic and non-zero for all  $z$  on a simple closed curve  $C$ .

Set  $\log f(z) = \ln|f(z)| + i \cdot \arg f(z)$

where a fixed branch for the logarithm is chosen. Then,

$$\int_C \frac{f'(z)}{f(z)} dz = \int_C d(\log f(z)) = \ln|f(z)| \Big|_C + i \cdot \arg f(z) \Big|_C$$

By the fundamental theorem of line integrals,  $\ln|f(z)| \Big|_C = 0$ , so

$$\int_C \frac{f'(z)}{f(z)} dz = i \cdot \arg f(z) \Big|_C$$

Let  $C'$  be the image of  $C$  under  $f$ . There are two cases;

Thm. Rouché Theorem.

Let  $f$  and  $g$  be analytic inside and on a simple closed contour  $C$ , with  $|g(z)| < |f(z)|$  for all  $z \in C$ .

Then,  $f+g$  and  $f$  have the same number of zeros inside  $C$ .

First proof. By the hypothesis,  $|g(z)| < |f(z)|$  on  $C$ . The triangle inequality gives

$$|f(z)| > 0 \quad \text{and} \quad |f(z) + g(z)| \geq |f(z)| - |g(z)| > 0 \quad (z \in C).$$

Therefore,  $f$  and  $f+g$  are analytic inside and on  $C$ , and have no zeros on  $C$ . By the analyticity, the number of poles,  $P_f = P_{f+g} = 0$ .

Since  $f$  and  $f+g$  cannot be zero on the contour  $C$ , write

$$f(z) + g(z) = f(z) \cdot \left(1 + \frac{g(z)}{f(z)}\right) = f(z) \cdot \phi(z) \quad \text{on } z \in C.$$

$$\begin{aligned} \text{Then, } (f+g)'(z) &= f'(z) \cdot \left(1 + \frac{g(z)}{f(z)}\right) + f(z) \cdot \left(1 + \frac{g(z)}{f(z)}\right)' \\ &= \frac{f'(z)}{f(z)} \cdot (f(z) + g(z)) + \left(1 + \frac{g(z)}{f(z)}\right)' \cdot (f(z) + g(z)) / \left(1 + \frac{g(z)}{f(z)}\right) \end{aligned}$$

$$\text{So that } \frac{(f+g)'(z)}{(f+g)(z)} = \frac{f'(z)}{f(z)} + \frac{\left(1 + \frac{g(z)}{f(z)}\right)'}{\left(1 + \frac{g(z)}{f(z)}\right)} = \frac{f'(z)}{f(z)} + \frac{\phi'(z)}{\phi(z)}.$$

Let  $N_{f+g}$  and  $N_f$  denote the number of zeros of  $f+g$  and  $f$ , respectively, on the domain, bounded by  $C$ . By the argument principle,

$$N_{f+g} - N_f = \frac{1}{2\pi i} \cdot \int_C \left[ \frac{(f+g)'(z)}{(f+g)(z)} - \frac{f'(z)}{f(z)} \right] dz = \frac{1}{2\pi i} \cdot \int_C \frac{\phi'(z)}{\phi(z)} dz. \quad (*)$$

Since  $|g(z)| < |f(z)|$ ,  $|g/f| < 1$  on  $C$ , so

$$|\phi(z) - 1| = |g(z)/f(z)| \leq 1.$$

Taking a branch of the logarithm containing  $D = \{w \in \mathbb{C} \mid |w-1| < 1\}$ , then

$$\frac{\phi'(z)}{\phi(z)} = \frac{d}{dz} \log \phi(z)$$

Therefore, the 'integral' in (\*) is zero, by the fundamental theorem of line integral, consequently  $N_{f+g} = N_f$ .  $\square$

Example.

Let  $P(z) = z^{11} + 3z^{10} + 6z^3 + 1$ . Determine the number of zeros of  $f$  in  $\{z \in \mathbb{C} \mid 1 < |z| < 2\}$ .

Proof. Let  $f(z) = 3z^{10}$  and  $g(z) = z^{11} + 6z^3 + 1$ . By the triangle inequality, on  $|z|=2$ ,

$$|g(z)| = |z^{11} + 6z^3 + 1| \leq |z|^{11} + 6|z|^3 + 1 \leq 2^{11} + 6 \cdot 2^3 + 1 \leq 3 \cdot 2^{10} = 2^{11} + 2^{10} = |f(z)|$$

Therefore, by the Rouché's Theorem,  $f+g$  has same number of zeros in  $|z| < 2$  as  $f$ , 10 zeros. And now, let  $f(z) = 6z^3$  and  $g(z) = z^{11} + 3z^{10} + 1$ . Observe that

$$|f(z)| = 6|z|^3 \quad \text{and} \quad |g(z)| = |z^{11} + 3z^{10} + 1| \leq |z|^{11} + 3|z|^{10} + 1$$

We can choose  $\delta > 0$  such that  $0 < \epsilon < \delta$  implies

$$|1 + \epsilon|^{11} + 3 \cdot |1 + \epsilon|^{10} + 1 < 6.$$

Therefore, for  $|z| = 1 + \epsilon$  where  $0 < \epsilon < \delta$ ,

$$|f(z)| = 6 \cdot |1 + \epsilon|^3 \geq |g(z)|,$$

So  $P(z)$  has three zeros inside  $|z| = 1 + \epsilon$ , for any  $0 < \epsilon < \delta$ .

Therefore, letting  $\epsilon \rightarrow 0$ ,  $P$  has three zeros on  $\{z \in \mathbb{C} \mid |z| \leq 1\}$ , consequently it has 7 zeros in the annulus  $1 < |z| < 2$ .  $\square$

✓ Evaluate

$$I = \int_{|z|=2} \frac{z+2}{z(z+1)} dz = \int_{|z|=2} \frac{\frac{1}{z^2} + \frac{2}{z^3}}{\frac{1}{z} + \frac{1}{z^2}} dz = - \int_{|z|=2} \frac{f(z)}{f'(z)} dz$$

where  $f(z) = \frac{1}{z} + \frac{1}{z^2} = \frac{z+1}{z^2}$ . It has order of 2 pole at  $z=0$ ,  
and has order of 1 zero at  $z=-1$ .

Therefore,  $I = -2\pi i \cdot (1-2) = 2\pi i$ .